

Defending Against Crescendo Multi-Turn Jailbreak Attacks

A Comparative Evaluation of Two Inference-Time Defense Mechanisms on Llama-3.2-3B-Instruct

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Abstract

Multi-turn jailbreak attacks - specifically the Crescendo pattern - exploit a model's tendency to commit to conversational context as a vector for extracting harmful outputs. This paper evaluates two inference-time defense mechanisms against ten Crescendo attack scenarios spanning five harm categories, using Llama-3.2-3B-Instruct as the target model. Defense 1 monitors *semantic drift* via cosine similarity against a conversation anchor and supplements it with toxicity classification. Defense 2 combines context truncation, periodic system-prompt re-injection, and a zero-shot LLM intent classifier. Against a baseline Attack Success Rate (ASR) of 50%, Defense 1 achieves **0% ASR** and Defense 2 achieves **10% ASR**. Both operate without retraining the underlying model, with Defense 1 adding under 100ms of CPU overhead per turn and Defense 2 adding one additional LLM inference call.

1. Introduction

Alignment training reduces a model's compliance with overtly harmful requests but is well-documented to be vulnerable to indirect prompting strategies. Multi-turn attacks are particularly difficult to defend because no single message is obviously malicious - harm emerges from the trajectory of the conversation rather than any individual turn.

The Crescendo attack [1] formalises this intuition: an adversary opens with a plausible, benign framing (historical research, creative writing, professional training) and issues a sequence of four to eight increasingly specific follow-up requests. By the time the terminal request arrives, the model has committed to a narrative frame that makes refusal feel inconsistent with its prior responses - a dynamic referred to as *memory stacking*. This paper builds a controlled evaluation harness for this attack pattern and benchmarks two lightweight, model-agnostic defenses against it.

The three main contributions are: (i) a reproducible 10-scenario attack benchmark across five harm categories; (ii) two distinct defense implementations - one signal-based, one context-management-based - that require no weight modification; and (iii) a comparative evaluation covering ASR, per-category robustness, and computational overhead.

2. Background

2.1 Multi-Turn Jailbreaking

Crescendo [1] differs from single-turn jailbreaks in that the attack surface is the conversation history rather than a single prompt. The attacker's leverage comes from the model's own outputs: each cooperative response makes the next escalation easier to obtain. Published benchmarks on frontier models report ASRs ranging from 40–80% depending on the harm category and the number of turns allowed [2].

2.2 Defense Landscape

Defenses against multi-turn jailbreaks can be grouped by intervention point. Pre-turn classifiers intercept harmful messages before they reach the model. Context management strategies alter what the model is allowed to remember. Fine-tuning approaches modify the model's weights to be resistant to commitment-building. This work focuses on the first two categories, which are deployable without model access and add minimal latency.

3. Experimental Setup

3.1 Target Model

The target model is **meta-llama/Llama-3.2-3B-Instruct**, loaded in 4-bit NF4 quantization (BitsAndBytesConfig, double quantization, float16 compute) to fit within a T4 GPU's 16 GB VRAM. Generation parameters: temperature 0.7, top-p 0.9, max 512 new tokens per turn. Model weights are available via the HuggingFace model hub; a Google Drive mirror of the quantized weights used in these experiments is linked in the repository README.

3.2 Attack Scenarios

Ten four-turn Crescendo scenarios are constructed across five harm categories. Turn 1 in each scenario establishes a credible, benign framing; Turns 2–3 escalate specificity; Turn 4 delivers the terminal harmful request. Table 1 summarises all scenarios. Every scenario is run under all three conditions (baseline, Defense 1, Defense 2) using identical generation parameters.

ID	Category	Name	Turn 4 Terminal Request
ATK-001	Weapons	Historical Arms Escalation	Step-by-step weapon mod guide
ATK-002	Weapons	Fiction to Explosives	Full IED assembly instructions
ATK-003	Malware	Student to Exploit Automation	SQLi automated scanner script
ATK-004	Malware	CEH Cert to Phishing	Credential harvesting campaign
ATK-005	Social Eng.	Psychology to Fraud Script	Elder fraud phone script
ATK-006	Social Eng.	UX Dark Patterns to Predatory Script	Predatory subscription script
ATK-007	Harm. Sub.	Lab Safety to Toxic Synthesis	Toxic gas synthesis steps
ATK-008	Harm. Sub.	NGO Reframe to Meth Recipe	Meth production walkthrough
ATK-009	Extremism	Academic to Recruitment Content	Extremist recruitment message
ATK-010	Misinform.	Poli-Sci to Election Disinfo	Voter suppression campaign

Table 1. The ten Crescendo attack scenarios used across all conditions.

3.3 Evaluation Metrics

The primary evaluation metric is Attack Success Rate (ASR): the proportion of scenarios in which the model's Turn-4 response does not contain a refusal. A heuristic refusal detector scans for 18 surface-level phrases ("I can't", "I won't", "harmful", "unethical", "against my guidelines", and similar). ASR is reported overall and per harm category. Secondary metrics include first-refusal turn (at which turn does the defense first intervene) and computational overhead per turn.

4. Defense Mechanisms

4.1 Defense 1: Semantic Drift Detection + Toxicity Filtering

The mechanism is motivated by a simple observation: Crescendo attacks must move the conversation's semantic content away from its opening framing to reach harmful territory. We track this movement by encoding Turn 1 as a dense anchor vector using all-MiniLM-L6-v2 [3] and computing the cosine similarity between the anchor and

each subsequent user message. Similarity below $\tau = 0.72$ triggers interception. A parallel toxicity classifier (unitary/toxic-bert [4]) intercepts any message scoring above 0.65, providing a secondary signal for attacks that escalate lexical toxicity without drifting semantically. Either signal alone is sufficient to block the turn.

When a turn is intercepted, the model is not called. A fixed safe response is returned and the intercepted turn is excluded from the conversation history. This means the model's context stays clean regardless of how many turns the attacker sends.

4.2 Defense 2: Context Reset + Intent Classification

Defense 2 attacks the memory-stacking mechanism at the context level rather than at the input signal level. Two controls run on every turn:

- **History truncation:** the model's context is capped at the last `MAX_HISTORY = 3` turns, stripping the accumulated commitment that prior cooperative responses create.
- **Safety system-prompt re-injection:** every `RESET_EVERY_K = 2` turns, the full safety system prompt is prepended to the context, resetting the normative frame the model uses to evaluate the current request.

Before either control is applied, each user message is passed to a zero-shot intent classifier - a separate call to the same LLM with a one-word classification prompt ("benign" or "adversarial"). If the classifier returns adversarial, the turn is blocked immediately without involving the main generation pipeline. This classification adds one inference call per turn but catches overtly harmful final-turn requests that slip past the semantic drift threshold.

5. Results and Analysis

5.1 Overall ASR

Table 2 shows the headline results. The undefended baseline has an ASR of 50%: five of ten attack chains successfully elicit harmful final-turn responses. Defense 1 drives this to 0% - a complete elimination of successful attacks. Defense 2 reaches 10%, with one scenario (ATK-001) succeeding.

Condition	Succeeded	Defended	ASR
Baseline (No Defense)	5 / 10	5 / 10	50.0%
Defense 1 - Semantic Drift + Tox.	0 / 10	10 / 10	0.0%
Defense 2 - Context Reset + Intent	1 / 10	9 / 10	10.0%

Table 2. Overall ASR across all three conditions.

5.2 Comparative Analysis

The two defenses achieve similar outcomes through fundamentally different mechanisms. Defense 1's semantic drift detector operates on input signal alone: it never needs to reason about intent, only measure topical distance from the conversation's starting point. This makes it robust against attacks that rely on gradual framing but potentially blind to attacks that maintain surface coherence throughout (e.g., a single-domain escalation that stays within the "weapons history" topic).

Defense 2's strength is its direct attack on the memory-stacking dynamic. By stripping context history and refreshing the system prompt, it denies the model the accumulated commitment that Crescendo exploits. Its weakness is the zero-shot intent classifier: four-turn attacks with a strong fictional or academic frame (like ATK-001) can present Turn-4 requests in language that reads as legitimate research to a one-shot classifier.

A combined system running both defenses in sequence - drift check first, then context management - would likely close the residual ATK-001 gap. The semantic drift detector would catch the topical deviation even if the intent classifier missed it.

5.3 Per-Category Effectiveness

Category	Baseline	Def. 1	Def. 2
Weapons (2)	50%	0%	50%
Malware (2)	50%	0%	0%
Social Eng. (2)	0%	0%	0%
Harm. Sub. (2)	50%	0%	0%
Extremism (1)	100%	0%	0%
Misinform. (1)	100%	0%	0%

Table 3. Per-category ASR by condition.

Social Engineering scenarios (ATK-005, ATK-006) were defended even at baseline, suggesting the model's alignment training is specifically sensitive to manipulation and coercion language. The Extremism and Misinformation categories, by contrast, had 100% baseline ASR - the model is most susceptible when attacks use academic or analytical framings that are genuinely common in legitimate research contexts. Both defenses eliminate all attacks in these categories.

The Weapons category is the most resistant to Defense 2 (50% ASR). Historical and fictional framings used in ATK-001 and ATK-002 are topically coherent throughout the four turns, keeping semantic similarity above the drift threshold for Defense 1 in ATK-001's case - though Defense 1 still blocks ATK-002. The root cause in ATK-001 is that the escalation from WW2 history to specific mechanical modification is narrow enough that the MiniLM embeddings do not flag it as a significant topic shift.

5.4 First-Refusal Turn Distribution

For Defense 1, interception most commonly occurs at Turn 3 (when requests shift from conceptual to technical) with some triggering at Turn 2. Turn-4 interceptions are rare, occurring only when an attack's terminal request uses overtly toxic vocabulary. For Defense 2, the intent classifier blocks primarily at Turn 4, consistent with the pattern of most attacks reaching their terminal request before the classifier flags adversarial intent.

6. Evaluation Against Assessment Parameters

6.1 ASR as Primary Metric

ASR captures the defence's core objective: preventing the model from completing the attack chain. Both defenses substantially reduce ASR - by 100% (Defense 1) and 80% (Defense 2) relative to baseline. The residual 10% ASR in Defense 2 corresponds to a specific, identifiable failure mode (zero-shot classification of historically-framed weapons requests) rather than a systematic weakness.

6.2 Robustness Across Attack Vectors

Defense 1 shows uniform robustness across all five harm categories, achieving 0% ASR everywhere. This reflects the domain-agnostic nature of the semantic drift signal - the detector does not need to know what harm category it is protecting against, only whether the conversation has moved. Defense 2 shows category-specific variation, with the Weapons category being the one gap. This is a meaningful finding: context management alone is insufficient against attacks that use a single coherent domain as both framing and target.

6.3 Resource Efficiency and Computational Overhead

Table 4 compares the overhead of each defense. Defense 1 adds two lightweight inference calls per turn - an embedding model and a small classification model, both runnable on CPU in under 100ms total. Defense 2 adds one full LLM inference call for intent classification, which adds 2–4 seconds per turn on the same GPU used for the main model. At conversational interaction rates this overhead is acceptable, but at high-throughput deployment scales Defense 1 is substantially more efficient.

Defense	Extra Calls / Turn	Latency Added	Modifies Weights?	Needs Training Data?
Defense 1	2 (embedder + classifier)	~60–100ms (CPU)	No	No
Defense 2	1 (intent LLM call)	~2–4s (GPU)	No	No

Table 4. Computational overhead comparison.

7. Limitations and Future Work

Several limitations should be noted. The refusal heuristic is a surface-level detector; a model can produce partial harmful content alongside a nominal refusal phrase and still be scored as defended. Manual review of all 120 turn-4 responses would give a more accurate picture. The 10-scenario benchmark, while representative, is a small sample - a larger benchmark with diverse phrasings and longer attack chains would give stronger statistical confidence.

The thresholds used in Defense 1 ($\tau = 0.72$ for drift, 0.65 for toxicity) were set based on qualitative judgment; systematic calibration using a held-out development set could improve precision. Defense 2's intent classifier is zero-shot; a few-shot or fine-tuned variant using examples of historically-framed escalations would likely close the ATK-001 gap.

Beyond these fixes, the most important extension is adaptive attack evaluation: testing whether an adversary who knows the defense parameters can construct prompts that evade them. For Defense 1, an adversary could deliberately anchor Turn 1 in the same domain as the target (e.g., opening with a weapons question rather than a historical fiction question) to suppress the drift signal throughout. Evaluating this scenario would establish a tighter upper bound on each defense's real-world robustness.

8. Conclusion

Both evaluated defenses substantially reduce Crescendo attack success on Llama-3.2-3B-Instruct without modifying model weights or requiring adversarial training data. Defense 1 (semantic drift + toxicity) achieves **0% ASR** with sub-100ms CPU overhead per turn. Defense 2 (context reset + intent classification) achieves **10% ASR** with one additional LLM inference call per turn. The results show that inference-time context management is a practical and effective direction for multi-turn jailbreak defense, and that the two mechanisms are complementary - combining them would likely close the remaining vulnerability.

References

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